

→ mathematical induction:

على ثلاث خطوات أساسية

STEP 1 → $n=1$

الخطوة الأولى

~ 2 → $n=k$

الخطوة الثانية

~ 3 → $n=k+1$

الخطوة الثالثة

Ex: $1+2+3+\dots+n = \frac{n(n+1)}{2}$

→ STEP 1 → $n=1$

$$1+2+3+\dots+1 = \frac{1(1+1)}{2}$$

$$L.H.S = 1$$

$$R.H.S = 1$$

→ STEP 2 → $n=k$

$$1+2+3+\dots+k = \frac{k(k+1)}{2}$$

→ STEP 3 → $n=k+1$

$$1+2+3+\dots+k+(k+1) = \frac{k+1(k+1+1)}{2}$$

$$\frac{k(k+1)}{2} + k+1 = \frac{k+1(k+2)}{2}$$

$$L.H.S = k+1 + \frac{k(k+1)}{2}$$

$$R.H.S = k+1 + \frac{(k+2)(k+1)}{2}$$

$$L.H.S \rightarrow \frac{2(k+1)}{2} + \frac{k(k+1)}{2}$$

$$\frac{(k+1)(2+k)}{2} \quad \text{①}$$

$$R.H.S \rightarrow \frac{2(k+1)}{2} + \frac{(k+1)(k+2)}{2} \quad \text{②}$$

هذه المتعادلة صحيحة

2) $n^3 + 2n$ divided by 3

→ ① $n = 1 \therefore 1^3 + 2 \times 1 = 3 \therefore \text{divided by } 3$

→ ② $n = k \therefore k^3 + 2k = 3M$

→ ③ $n = k+1 \therefore (k+1)^3 + 2(k+1)$

~~$k^3 + 3k^2 + 3k + 1 + 2k + 2$~~

~~$(k^3 + 2k) + 3k^2 + 3k + 3$~~

$\therefore k^3 + 3k^2 + 3k + 1 + 2k + 2$

$= (k^3 + 2k) + 3k^2 + 3k + 3$

$= 3M + 3k^2 + 3k + 3$

$3(M + k^2 + k + 1)$

$P = (M + k^2 + k + 1)$ هتفر من اوان

$3P \therefore 3 \times \text{عدد صحيح}$

$(k+1)^3 + 2(k+1) \therefore$

$3 \therefore \text{قيل المقام}$

partial fractions:

الـ partial fraction هو عليه نكتب لتوحيد المقامات
شروط صوب التفسير الخريف:

- 1- درجة المقام < درجة البسط
- 2- المقام ميكونش فيه $(\sqrt{\quad})$

1 case: DISTINCT LINEAR factors.

المقام من الدرجة الأولى
Resolve into partial fraction

$$1) f(x) = \frac{7x - 25}{(x-3)(x-4)}$$

solution: $\frac{7x - 25}{(x-3)(x-4)} = \frac{A}{(x-3)} + \frac{B}{(x-4)}$

صوب مطلق
كلنا
المقام من الدرجة الأولى

← (مقرب البعد له المقام قبل القاء)

$$\left[\frac{7x - 25}{(x-3)(x-4)} = \frac{A}{(x-3)} + \frac{B}{(x-4)} \right] \times (x-3)(x-4)$$

$$\rightarrow (7x - 25) = A(x-4) + B(x-3)$$

← (عشان أجيب A بنزله اظلم مع B)

A → «x=3» $\therefore 7 \times 3 - 25 = A(3-4) + B(3-3)$

$$\therefore -4 = -A \quad \boxed{\therefore A = 4}$$

← (عشان أجيب B بنزله اظلم مع A)

B → «x=4» $\therefore 7 \times 4 - 25 = A(4-4) + B(4-3)$

$$\therefore 3 = B \quad \boxed{3 = B}$$

$$\therefore f(x) = \frac{4}{(x-3)} + \frac{3}{(x-4)}$$

$$F_{x_2} \rightarrow f(x) = \frac{1}{(x-1)(x+2)(x+3)}$$

solution:

$$\frac{1}{(x-1)(x+2)(x+3)} = \frac{A}{(x-1)} + \frac{B}{(x+2)} + \frac{C}{(x+3)}$$

$$\left[\frac{1}{(x-1)(x+2)(x+3)} = \frac{A}{(x-1)} + \frac{B}{(x+2)} + \frac{C}{(x+3)} \right] \times \frac{(x-1)(x+2)}{(x+3)} \leftarrow$$

$$\rightarrow 1 = A(x+2)(x+3) + B(x-1)(x+3) + C(x-1)(x+2)$$

$$\rightarrow A = \text{("x=1")}$$

$$1 = A(1+2)(1+3) + B(1-1)(1+3) + C(1-1)(1+2)$$

$$\therefore 1 = 12A \quad \therefore \boxed{A = \frac{1}{12}}$$

$$\rightarrow B = \text{("x=-2")}$$

$$1 = A(-2+2)(-2+3) + B(-2-1)(-2+3) + C(-2-1)(-2+3)$$

$$\therefore 1 = -3B \quad \therefore \boxed{B = -\frac{1}{3}}$$

$$\rightarrow C = \text{("x=-3")}$$

$$\therefore 1 = A(-3+2)(-3+3) + B(-3-1)(-3+3) + C(-3-1)(-3+2)$$

$$\therefore 1 = 4C \quad \therefore \boxed{C = \frac{1}{4}}$$

$$\therefore f(x) = \frac{\frac{1}{12}}{(x-1)} + \frac{-\frac{1}{3}}{(x+2)} + \frac{\frac{1}{4}}{(x+3)}$$

Case 2 → Repeated linear factors...

← يكون في المقام قوس متكرر أو متكرر (الذي متكرر) ←
 ← Resolve into partial fraction: ←

Ex1: Resolve into partial fraction:

$$1) \frac{1}{x(x+2)^2}$$

solution:

$$\frac{1}{x(x+2)^2} = \frac{A}{x} + \frac{B}{(x+2)} + \frac{C}{(x+2)^2}$$

$$\rightarrow \left[\frac{1}{x(x+2)^2} = \frac{A}{x} + \frac{B}{(x+2)} + \frac{C}{(x+2)^2} \right]$$

$$\rightarrow 1 = A(x+2)^2 + Bx(x+2) + Cx$$

$$\rightarrow A \rightarrow ("x=0")$$

$$\therefore 1 = A(0+2)^2 + B(0)(0+2) + C(0)$$

$$\therefore 1 = 4A \quad \left\{ A = 0.25 = \frac{1}{4} \right\}$$

$$\rightarrow B \rightarrow ("x = -2")$$

$$1 = A(x^2 + 4x + 4) + B(x^2 + 2x) + Cx$$

$$x^2: 0 = A + B \quad \therefore B = -A \quad \therefore B = -\frac{1}{4}$$

$$\rightarrow C \rightarrow ("x = -2")$$

$$\therefore 1 = A(-2+2)^2 + B(-2)(-2+2) + C(-2)$$

$$1 = -2C \quad \therefore C = -\frac{1}{2}$$

$$f(x) = \frac{1/4}{x} + \frac{-1/4}{(x+2)} + \frac{-1/2}{(x+2)^2}$$

$$Ex2: \frac{x^2 - 3x + 1}{(x-1)^2(x-2)}$$

solutions:

$$\frac{x^2 - 3x + 1}{(x-1)^2(x-2)} = \frac{A}{(x-2)} + \frac{B}{(x-1)} + \frac{C}{(x-1)^2}$$

$$\rightarrow \left[\frac{x^2 - 3x + 1}{(x-1)^2(x-2)} = \frac{A}{(x-2)} + \frac{B}{(x-1)} + \frac{C}{(x-1)^2} \right]$$

$$\times (x-1)^2(x-2)$$

$$\rightarrow \therefore x^2 - 3x + 1 = A(x-1)^2 + B(x-1)(x-2) + C(x-2)$$

$$A \rightarrow ((x=2))$$

$$2^2 - 3 \cdot 2 + 1 = A(2-1)^2 + B(2-1)(2-2) + C(2-2)$$

$$\boxed{-1 = A}$$

$$B \rightarrow$$

$$A(x^2 - 2x + 1) + B(x^2 - 3x + 2) + C(x-2)$$

$$x^2 - 1 = A + B \quad \therefore B = \boxed{1 - A}$$

$$\boxed{\therefore B = 2}$$

$$C \rightarrow ((x=1))$$

$$1^2 - 3 \cdot 1 + 1 = A(1-1)^2 + B(1-1)(1-2) + C(1-2)$$

$$-1 = -C \quad \boxed{C = 1}$$

$$L(x) = \frac{-1}{x-2} + \frac{2}{(x-1)} + \frac{1}{(x-1)^2}$$

$$F_{x3} \rightarrow \frac{1}{x^4(x+1)}$$

solution:

$$\frac{1}{x^4(x+1)} = \frac{A}{(x+1)} + \frac{B}{x} + \frac{C}{x^2} + \frac{D}{x^3} + \frac{E}{x^4}$$

$$\left[\frac{1}{x^4(x+1)} = \frac{A}{(x+1)} + \frac{B}{x} + \frac{C}{x^2} + \frac{D}{x^3} + \frac{E}{x^4} \right] \times x^4(x+1)$$

$$\rightarrow 1 = A(x^4) + Bx^3(x+1) + Cx^2(x+1) + Dx(x+1) + E(x+1)$$

$$\rightarrow A \rightarrow A(x^4) + B(x^4 + x^3) + C(x^3 + x^2) + D(x^2 + x) + E(x+1)$$

$$x^4 = 0 = A + B \quad \therefore A = -B$$

$$\therefore A = 1$$

$$\rightarrow B \rightarrow x^3 = 0 = B + C \quad \therefore B = -C$$

$$\therefore B = -1$$

$$\rightarrow C \rightarrow x^2 = 0 = C + D \quad \therefore C = -D \quad \therefore C = 1$$

$$\rightarrow D \rightarrow x = 0 = D + E \quad \therefore D = -E$$

$$\therefore D = -1$$

$$\rightarrow E \rightarrow (x=0)$$

$$1 = A(0^4) + B(0^3)(0+1) + C(0^2)(0+1) + D(0)(0+1) + E(0+1)$$

$$\therefore 1 = E$$

case 3

distinct linear and quadratic factors:-

بعض اند هيدنيا قوسين في القالب
واحد مستطال وواحد مربع

$$\frac{9x-7}{(x+3)(x^2+1)}$$

solution:

$$\frac{9x-7}{(x+3)(x^2+1)} = \frac{A}{(x+3)} + \frac{Bx+C}{(x^2+1)} \quad \text{[لبنه]}$$

$$\left[\frac{9x-7}{(x+3)(x^2+1)} = \frac{A}{(x+3)} + \frac{Bx+C}{(x^2+1)} \right] \times (x+3)(x^2+1)$$

$$\rightarrow 9x-7 = A(x^2+1) + (Bx+C)(x+3)$$

$$A \rightarrow \text{"} x = -3 \text{"}$$

$$9(-3)-7 = A(-3^2+1) + (B(-3)+C)(-3+3)$$

$$-34 = 10A \quad \therefore A = -3.4$$

$$B, C \rightarrow \text{"} x =$$

(Léo, ni, Léo)

$$\rightarrow 9x-7 = Ax^2+A+Bx^2+3Bx+Cx+3C$$

$$x^2: 0 = A+B \quad \therefore B = -A \quad \therefore B = 3.4$$

$$x: 9 = 3B+C$$

$$\therefore 9 = (3 \times 3.4) + C$$

$$C = 9 - 10.2$$

$$\therefore C = -\frac{6}{5}$$

$$\begin{aligned} & \cancel{9 = 10.2 + C} \\ & \cancel{\therefore C = 15.8} \\ & \cancel{18} \end{aligned}$$

Case 4 Repeated Quadratic factors

$$F_x = \frac{8x^3 + 13x}{(x^2 + 2)^2}$$

solution:

$$\left[\frac{8x^3 + 13x}{(x^2 + 2)^2} = \frac{Ax + B}{(x^2 + 2)} + \frac{Cx + D}{(x^2 + 2)^2} \right] \times (x^2 + 2)^2$$

$$\rightarrow 8x^3 + 13x = Ax + B(x^2 + 2) + Cx + D$$

$$\begin{aligned} \rightarrow 8x^3 + 13x &= Ax^3 + 2Ax + Bx^2 + 2B + Cx + D \\ &= Ax^3 + (2A + C)x + Bx^2 + (2B + D) \end{aligned}$$

$$A = 8$$

$$2A + C = 13$$

$$16 + C = 13$$

$$D = 0$$

$$\begin{aligned} C &= -3 \\ B &= 0 \end{aligned}$$